STUDY SESSION / OCTOBER 27, 2026

Linear Programming in R

Session record: formulation, constraints, shortest paths, binary variables, and solution interpretation

Covered problems

- Question 1 - Move-It Company (forklift shipping)
- Question 2 - New York to Los Angeles (minimum-fuel route)
- Question 3 - Jane Driver (car maintenance and replacement)

How to use this PDF

This is a formatted record of the questions, code, outputs, explanations, and corrections from the session. User-supplied R scripts are retained with only line breaks adjusted for page readability. It is a study reference, not a claim that the assistant executed the R scripts.

Quick answers: **$34,000** weekly shipping cost | **2,450 gallons** shortest route | **$31,000** maintenance-plan cost

Session at a glance

Topic	Decision variables	Answer / key point
Q1: Move-It	6 shipment quantities: XA1 ... XB3	Minimum $34,000; Plant A ships to centers 2 and 3, Plant B to center 1
Q2: Road route	14 directed arc choices	Minimum 2,450 gallons; A -> C -> E -> H
Binary clarification	0 = road not used; 1 = road used	Binary is intuitive, but flow-balance LP already has an integral optimum
Q3: Jane Driver	15 year-pair car-replacement choices	Minimum $31,000; X13, X34 and X46 are 1

The sections below follow the conversation in sequence. Each contains the user's formulation or code, the explanation given, and the final interpretation.

Question 1 - Move-It Company

USER PROBLEM STATEMENT

Two plants, A and B, make forklift trucks for three distribution centers. Sixty trucks are produced and shipped weekly. Each plant can ship at most 50 trucks. Each distribution center must receive exactly 20 trucks. Production costs are identical; the goal is to minimize shipping costs.

Plant	Center 1	Center 2	Center 3
A	$800	$700	$400
B	$600	$800	$500

1. Decision variables (user formulation)

XA1, XA2, XA3 = forklifts shipped from Plant A to distribution centers 1, 2, and 3.

XB1, XB2, XB3 = forklifts shipped from Plant B to distribution centers 1, 2, and 3.

2. Objective function (user formulation)

Minimize Z = 800 XA1 + 700 XA2 + 400 XA3 + 600 XB1 + 800 XB2 + 500 XB3

3. Constraints (user formulation)

```
Plant A:  XA1 + XA2 + XA3 <= 50
Plant B:  XB1 + XB2 + XB3 <= 50
Center 1: XA1 + XB1 = 20
Center 2: XA2 + XB2 = 20
Center 3: XA3 + XB3 = 20
Total:    XA1 + XA2 + XA3 + XB1 + XB2 + XB3 = 60
Nonnegativity: XA1, XA2, XA3, XB1, XB2, XB3 >= 0
```

Initial question: how do the constraints become R vectors?

The user started with `make.lp(0,3)` and an example constraint `c(0.25,0.25)`. The explanation was to use six columns, because there are six decision variables, and to keep their positions consistent.

```
ProductMix.LP <- make.lp(0,6)
# Column order: XA1, XA2, XA3, XB1, XB2, XB3
add.constraint(ProductMix.LP, c(1,1,1,0,0,0), "<=", 50)
```

The coefficient vector `c(1,1,1,0,0,0)` refers only to Plant A. The corresponding vector for Plant B is `c(0,0,0,1,1,1)`. For Center 1, use `c(1,0,0,1,0,0)`, since XA1 and XB1 are the two ways trucks can arrive.

Complete R code subsequently submitted by user

```
library(lpSolve)
library(lpSolveAPI)

ProductMix.LP <- make.lp(0,6)
add.constraint(ProductMix.LP,c(1,1,1,0,0,0), "<=", 50)
add.constraint(ProductMix.LP,c(0,0,0,1,1,1), "<=", 50)
add.constraint(ProductMix.LP,c(1,0,0,1,0,0), "=", 20)
add.constraint(ProductMix.LP,c(0,1,0,0,1,0), "=", 20)
add.constraint(ProductMix.LP,c(0,0,1,0,0,1), "=", 20)
add.constraint(ProductMix.LP,c(1,1,1,1,1,1), "=", 60)
#set.bounds(ProductMix.LP,upper=c(500,500),lower=c(-100000,-100000),c(1,2))
set.objfn(ProductMix.LP,c(800,700,400,600,800,500))
lp.control(ProductMix.LP,sense="min")
solve(ProductMix.LP)
write.lp(ProductMix.LP, filename="ProductMixLP.lp")
print(ProductMix.LP)

# optimal objective function
get.objective(ProductMix.LP)
# optimal
get.variables(ProductMix.LP)
```

ASSISTANT VERIFICATION

All six constraints matched the formulation. The objective coefficients and minimization direction were correct. The total-shipped constraint is redundant because the three center requirements already sum to 60; it can remain. The commented bounds line is from another example and can be removed. The default variable lower bound is zero.

Variable	Optimal value	Meaning
XA1	0	No shipments from A to center 1
XA2	20	A to center 2
XA3	20	A to center 3
XB1	20	B to center 1
XB2	0	No shipments from B to center 2
XB3	0	No shipments from B to center 3

Question 1 result

Minimum weekly shipping cost = **$34,000**. Plant A ships 40 trucks; Plant B ships 20 trucks. Each distribution center receives 20 trucks.

Question 2 - New York to Los Angeles

USER PROBLEM STATEMENT

A truck must travel from New York to Los Angeles along directed arcs. Each arc consumes a specified number of gallons. Find the route with the fewest total gallons.

A = New York; B = Cleveland; C = St. Louis; D = Nashville; E = Phoenix; F = Dallas; G = Salt Lake City; H = Los Angeles.

1. Decision variables

For each directed arc, Xij = 1 if the truck uses the arc from i to j, and 0 otherwise. The model has 14 directed arcs.

Position	Arc	Gallons
1	XAB	400
2	XAC	950
3	XAD	800
4	XBE	1,800
5	XBF	900
6	XCE	1,100
7	XCF	600
8	XDF	600
9	XDG	1,200
10	XEH	400
11	XFE	900
12	XFG	1,000
13	XFH	1,300
14	XGH	600

2. Objective function

The user wrote an objective containing the costs of the arcs, but inadvertently omitted 1,800 XBE in the written expression. The R code included that term correctly. The complete objective is:

```
Minimize Z =
400 XAB + 950 XAC + 800 XAD + 1800 XBE + 900 XBF
+ 1100 XCE + 600 XCF + 600 XDF + 1200 XDG
+ 400 XEH + 900 XFE + 1000 XFG + 1300 XFH + 600 XGH
```

3. Flow-balance constraints (user formulation)

```
New York:     XAB + XAC + XAD = 1
Cleveland:    XAB = XBE + XBF
St. Louis:    XAC = XCE + XCF
Nashville:    XAD = XDF + XDG
Phoenix:      XBE + XCE + XFE = XEH
Dallas:       XBF + XCF + XDF = XFE + XFG + XFH
Salt Lake:    XDG + XFG = XGH
Los Angeles:  XEH + XFH + XGH = 1
```

The original written Salt Lake City line said "XFEG"; the intended variable is XFG. The submitted R vector used XFG correctly.

R code submitted by user

```
library(lpSolve)
library(lpSolveAPI)

ProductMix.LP <- make.lp(0,14)
add.constraint(ProductMix.LP,c(1,1,1,0,0,0,0,0,0,0,0,0,0,0), "=", 1)
add.constraint(ProductMix.LP,c(1,0,0,-1,-1,0,0,0,0,0,0,0,0,0), "=", 0)
add.constraint(ProductMix.LP,c(0,1,0,0,0,-1,-1,0,0,0,0,0,0,0), "=", 0)
add.constraint(ProductMix.LP,c(0,0,1,0,0,0,0,-1,-1,0,0,0,0,0), "=", 0)
add.constraint(ProductMix.LP,c(0,0,0,1,0,1,0,0,0,-1,1,0,0,0), "=", 0)
add.constraint(ProductMix.LP,c(0,0,0,0,1,0,1,1,0,0,-1,-1,-1,0), "=", 0)
add.constraint(ProductMix.LP,c(0,0,0,0,0,0,0,0,1,0,0,1,0,-1), "=", 0)
add.constraint(ProductMix.LP,c(0,0,0,0,0,0,0,0,0,1,0,0,1,1), "=", 1)
#set.bounds(ProductMix.LP,upper=c(500,500),lower=c(-100000,-100000),c(1,2))
set.objfn(ProductMix.LP,c(400,950,800,1800,900,1100,600,
                          600,1200,400,900,1000,1300,600))
lp.control(ProductMix.LP,sense="min")
solve(ProductMix.LP)
write.lp(ProductMix.LP, filename="ProductMixLP.lp")
print(ProductMix.LP)

# optimal objective function
get.objective(ProductMix.LP)
# optimal
get.variables(ProductMix.LP)
```

Assistant verification and outcome

All eight balance constraints in R were consistent with the directed arcs. The objective vector was correct. The main corrections were to the written objective (add 1800 XBE) and written Salt Lake City constraint (XFG rather than XFEG).

```
get.objective(ProductMix.LP)
[1] 2450

get.variables(ProductMix.LP)
[1] 0 1 0 0 0 1 0 0 0 1 0 0 0 0
```

Question 2 result

Selected variables: **XAC = 1, XCE = 1, XEH = 1**, all others 0. Route: **New York -> St. Louis -> Phoenix -> Los Angeles**. Fuel used: **950 + 1,100 + 400 = 2,450 gallons**.

Follow-up - Why use binary variables?

USER QUESTION

"Can you help me understand why you would do the binary?" The user also shared the Question 2 output of 2,450 and the 14-element zero/one vector.

Assistant explanation

Binary means a variable can only be 0 or 1. In the route model, 0 means that arc is not used and 1 means that arc is used. Standard LP variables are continuous by default; a fractional value could be interpreted as splitting flow across different paths, even though one truck should follow one route.

```
# Optional: explicitly constrain all 14 arcs to be binary
set.type(ProductMix.LP, columns=1:14, type="binary")
```

However, for this standard shortest-path formulation, the flow-balance constraint matrix has a structure that guarantees an integral optimal extreme-point solution with integer supplies and demands. Therefore, **binary restrictions are not necessary to obtain the correct shortest path in this case**. The user's existing LP already returned 0s and 1s.

Binary restrictions can be useful to express the meaning of a route choice, and can become important if additional constraints break the usual shortest-path structure.

Question 3 - Jane Driver's Car Maintenance

USER PROBLEM STATEMENT

Jane Driver must choose when to replace her car over a five-year planning period. Represent a car purchase and its next replacement with an arc from year i to year j. The goal is to minimize total cost of the selected replacement schedule.

1. Decision variables (user formulation, refined)

Xij = 1 if Jane buys a car at the beginning of year i and replaces it at the beginning of year j; Xij = 0 otherwise. When j = 6, she keeps the car to the end of Year 5.

Year node	Meaning
1	Beginning of Year 1 (initial purchase)
2	Beginning of Year 2 (possible replacement)
3	Beginning of Year 3 (possible replacement)
4	Beginning of Year 4 (possible replacement)
5	Beginning of Year 5 (possible replacement)
6	End of Year 5 (end of planning period)

2. Objective function (user formulation)

```
Minimize Z =
7000 X12 + 12000 X13 + 21000 X14 + 31000 X15 + 44000 X16
+ 7000 X23 + 12000 X24 + 21000 X25 + 31000 X26
+ 7000 X34 + 12000 X35 + 21000 X36
+ 7000 X45 + 12000 X46 + 7000 X56
```

3. Constraints (user formulation)

```
Year 1: X12 + X13 + X14 + X15 + X16 = 1
Year 2: X12 = X23 + X24 + X25 + X26
Year 3: X13 + X23 = X34 + X35 + X36
Year 4: X14 + X24 + X34 = X45 + X46
Year 5: X15 + X25 + X35 + X45 = X56
Year 6: X16 + X26 + X36 + X46 + X56 = 1
```

R code submitted by user

```
library(lpSolve)
library(lpSolveAPI)

ProductMix.LP <- make.lp(0,15)
add.constraint(ProductMix.LP,c(1,1,1,1,1,0,0,0,0,0,0,0,0,0,0), "=", 1)
add.constraint(ProductMix.LP,c(1,0,0,0,0,-1,-1,-1,-1,0,0,0,0,0,0), "=", 0)
add.constraint(ProductMix.LP,c(0,1,0,0,0,1,0,0,0,-1,-1,-1,0,0,0), "=", 0)
add.constraint(ProductMix.LP,c(0,0,1,0,0,0,1,0,0,1,0,0,-1,-1,0), "=", 0)
add.constraint(ProductMix.LP,c(0,0,0,1,0,0,0,1,0,0,1,0,1,0,-1), "=", 0)
add.constraint(ProductMix.LP,c(0,0,0,0,1,0,0,0,1,0,0,1,0,1,1), "=", 1)
#set.bounds(ProductMix.LP,upper=c(500,500),lower=c(-100000,-100000),c(1,2))
set.objfn(ProductMix.LP,c(7000,12000,21000,31000,44000,
                          7000,12000,21000,31000,
                          7000,12000,21000,
                          7000,12000,7000))
set.type(ProductMix.LP, columns=1:14, type="binary")
lp.control(ProductMix.LP,sense="min")
solve(ProductMix.LP)
write.lp(ProductMix.LP, filename="ProductMixLP.lp")
print(ProductMix.LP)

# optimal objective function
get.objective(ProductMix.LP)
# optimal
get.variables(ProductMix.LP)
```

Output provided by user

```
get.objective(ProductMix.LP)
[1] 31000

get.variables(ProductMix.LP)
[1] 0 1 0 0 0 0 0 0 0 1 0 0 0 1 0
```

Assistant verification and correction

The model has 15 decision variables. All six flow-balance constraints and the 15 objective coefficients are correctly aligned with the listed arcs. One small coding correction: the submitted binary specification only covers columns 1 through 14, leaving X56 continuous. If you are explicitly making every variable binary, use:

```
set.type(ProductMix.LP, columns=1:15, type="binary")
```

The final zero/one result is nevertheless valid. Binary restrictions are not strictly needed for a standard shortest-path network formulation, but specifying all 15 expresses the decision-variable definition consistently.

Selected arc	Time period	Cost
X13 = 1	Purchase in Year 1; replace in Year 3	$12,000
X34 = 1	Purchase in Year 3; replace in Year 4	$7,000
X46 = 1	Purchase in Year 4; keep until end of Year 5	$12,000
Total	1 -> 3 -> 4 -> 6	$31,000

Question 3 result

Minimum modeled cost = **$31,000**. The selected replacement sequence is **1 -> 3 -> 4 -> 6**. More than one optimal path can have the same total cost; this is one valid optimum.

Practical takeaways from the session

Column order is everything.

Each add.constraint coefficient must correspond to the same decision variable in every row and in set.objfn.

Flow balance is incoming equals outgoing.

At the start node, one unit leaves; at intermediate nodes, what comes in goes out; at the destination, one unit arrives.

Continuous versus binary.

Quantities shipped can be continuous/nonnegative; a yes/no route decision is conceptually binary. Standard shortest-path LPs often solve integrally without explicit binary types.

Interpret the variable vector.

Use the predefined arc order to translate the 1s in get.variables() into an actual path or replacement schedule.

Check small discrepancies.

Question 2 had two written typos; Question 3 needed columns=1:15 if all car-replacement arcs were to be declared binary.